

Day 3 Homework: Choosing and proving a model

Mithran Sadasivam · AI Engineer Program, Phase 1 · Code and data: github.com/mithransadasivam/FDE
(branch hw3, folder "HW #3")

1. Memo: MODEL_CHOICE

Data: 10 prompts × 3 models = 30 calls, one run, reply cap 600 tokens. Quality scores (1–5) were assigned by Claude, not by a human reviewer, on correctness, following the instruction, and format. The prompts are the ten course IT prompts (no names or customer data), not prompts from my own work.

Recommendation

Use hosted (claude-haiku-4.5) as the default for classification, extraction, summaries, commit messages and test-case drafts. Escalate to strong (claude-sonnet-4.5) for anything that needs reasoning about a rule or trade-off (the batch-window prompt is the one clear case in this data), and keep local only as the fallback when the hosted provider is unreachable.

Evidence

Backend	Model	Avg score (1–5)	Avg seconds	Cost, 10 prompts	Cost per call
hosted	anthropic/claude-haiku-4.5	4.1	1.89	\$0.0057	\$0.00057
strong	anthropic/claude-sonnet-4.5	4.5	3.83	\$0.0187	\$0.00187
local	llama3.2:3b (Ollama)	2.8	7.97 (5.48 without the 30.4 s first call)	\$0	\$0

- strong beat hosted by 0.4 points on average. Its clearest win was prompt 4 (the 05:50 batch job that must not run after 06:00): strong 5 vs hosted 3. Both said "do not start", but hosted's reason was wrong (it treated finishing at 06:15 as before the 06:30 report). Local said the job could run: 1.
- The two hosted models tied on prompt 3 (JSON extraction, 5 each) and were within 1 point on prompts 1, 2, 6, 7, 8 and 9.
- hosted scored higher than strong on prompt 5 (customer rewrite, 4 vs 3: strong kept "As mentioned previously") and prompt 10 (test cases, 5 vs 3). Prompt 10 is partly an artefact: strong's answer hit the 600-token cap and was cut off mid-step.
- local scored 3 or below on 7 of 10 prompts: it added unrequested fields on prompt 3, chatted around the answer on prompt 7, and wrote an incoherent test plan on prompt 10.
- Speed: hosted fastest (mean 1.89 s); strong about twice as slow (3.83 s).

Limits: 10 prompts, one run each, one scorer (not the user), and a small quality gap between the hosted models. This supports "use the cheap model by default", not a precise ranking.

Cost

Per-call costs from the benchmark, projected at an assumed 1,000 calls a day for 30 days (real team volume not measured):

Plan	Monthly cost
All calls on hosted	about \$17
All calls on strong	about \$56
hosted default, 20% escalated to strong	about \$25
local	\$0 in API fees (own hardware and time)

The cost report in section 2 (longer chat replies) gives higher figures: \$35 and \$95 a month. Output tokens were 76% of the cost there, so asking for shorter answers may save as much as switching models.

Risks

- **Privacy:** prompts sent to hosted and strong leave the machine through OpenRouter. Redact names, customer data and internal identifiers first; anything that cannot be redacted goes to local, whose quality here was the weakest.
- **Rate limits and credit:** OpenRouter returned HTTP 402 during this work because the account balance was too low for the default 64,000-token reply limit. The benchmark and the chatbot now send a max_tokens cap, and ask() retries rate limits and timeouts three times with a doubling wait.
- **Provider down:** ask() falls back to the local backend after the retries fail. Local needs Ollama running with llama3.2:3b pulled, and its answers were the weakest, so treat fallback answers as lower confidence.

Review

Repeat the benchmark on ten of my own work prompts, scored by a person, by 2026-12-28, or sooner if OpenRouter prices change or a new model is released. The decision would change if the score gap between hosted and strong grows on real prompts, if the escalation rate goes well above 20%, or if local reaches a score of 4 on the tasks I need it for.

Routing rule added to CLAUDE.md: default to the hosted backend; escalate to strong only for reasoning about rules or trade-offs, or text a customer will read, and say why in the commit message; local is the fallback and for data that must not leave the machine.

2. Cost report (output of scripts/cost_report.py)

Run on data/usage.csv: 9 calls, three per backend, each a 3-message chat through the chatbot. Hand check: first haiku row = 31 input × \$1/1M + 202 output × \$5/1M = \$0.001041, equal to the logged cost.

model	calls	in_tok	out_tok	cost	\$/call
anthropic/claude-haiku-4.5	3	821	530	\$ 0.0035	\$ 0.00116
anthropic/claude-sonnet-4.5	3	749	486	\$ 0.0095	\$ 0.00318
llama3.2:3b	3	526	281	\$ 0.0000	\$ 0.00000

output tokens are 76% of total cost

projection at 1,000 calls/day (30 days):

anthropic/claude-haiku-4.5	\$34.71/month
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anthropic/claude-sonnet-4.5	\$95.37/month
llama3.2:3b	\$0.00/month

3. Git log (git log --oneline, branch hw3)

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53d6c97 HW #3 Task 4: score benchmark, add model choice memo and routing rule
8f4ca6b HW #3 Task 3: add cost report, usage log and reply cap in chatbot
c7708e3 HW #3 Task 2: cap reply length in benchmark
54977bf HW #3 Task 2: add benchmark script, tests and prompt file
d28a796 HW #3 Task 1: build model registry from .env and add a third backend
```

Task 1 is one commit, Task 3 is one commit and Task 4 is one commit. Task 2 has two: the script, then a follow-up that capped reply length after OpenRouter returned HTTP 402.

4. Reflection

I would put the hosted model, claude-haiku-4.5, in front of my team as the default, with claude-sonnet-4.5 behind it for questions that need reasoning about a rule. Haiku is twice as fast and about a third of the price, and it was within a point of Sonnet on most of the ten prompts.

The result that changed my mind was prompt 4, the batch job that must not run after 06:00. Before the benchmark I expected the cheap model to be fine for any short question. Haiku did give the right answer (do not start) but for the wrong reason, saying the job finishing at 06:15 was safely before the 06:30 report, when the rule is about 06:00. A wrong reason behind a right answer is the kind of error that would slip past a quick review, and it is why I would route rule-based questions to Sonnet rather than only looking at answers. The local model told me to start the job, which would have broken the rule.

Caveats: the quality scores were assigned by Claude and should be re-scored by a person, and the prompts are the course prompts rather than my own work.